南方科技大学 SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

考试科目: 高等数学(下) A 开课单位: 数 学 系

考试时长: 120 分钟 命题教师:

题 号	1	2	3	4	5	6	7	8	9
分 值	15 分	15 分	10 分	10 分	10 分	10 分	10 分	10 分	10 分

本试卷共 9 大题, 满分 100 分. (考试结束后请将试卷、答题本、草稿纸一起交给监考老师)

注意：本试卷里的中文为直译（即完全按英文字面意思直接翻译），所有数学词汇的定义请参照教材（Thomas' Calculus，13th Edition）中的定义。如果其中有些数学词汇的定义不同于中文书籍（比方说同济大学的高等数学教材）里的定义，以教材（Thomas' Calculus，13th Edition）中的定义为准。

1. (15 pts) **Multiple Choice Questions:** (only one correct answer for each of the following questions.)

   (1) The interval of convergence for the power series $\sum_{n=1}^{\infty} \frac{x^n}{n3^n}$ is

   (A) $[-1/3, 1/3]$. (B) $[-1/3, 1/3)$.

   (C) $[-3, 3]$. (D) $[-3, 3)$.

   (2) Let $f(x,y) = \begin{cases} y^2 \sin \frac{1}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$. Which of the following statements is **wrong** ?

   (A) $f(x,y)$ is continuous at $(0,0)$.

   (B) $f_x(0,0)$ exists.

   (C) $f_y(0,0)$ exists.

   (D) $f_x(x,y)$ is continuous at $(0,0)$.

   (3) If $f(x,y)$ has partial derivatives at (x_0, y_0), then

   (A) $f(x,y)$ is bounded around (x_0, y_0).

   (B) $f(x,y)$ is continuous around (x_0, y_0).

   (C) $f(x, y_0)$ is continuous at x_0, $f(x_0, y)$ is continuous at y_0.

   (D) $f(x,y)$ is continuous at (x_0, y_0).

   (4) Let a be a constant. Then the series $\sum_{n=1}^{\infty} \left(\frac{\sin(an)}{n^2} + \frac{(-1)^n}{n+1} \right)$

   (A) converges absolutely.

   (B) converges conditionally.

(C) diverges.

(D) the convergence depends on the value of a.

(5) $\int_0^1 \int_y^1 \frac{\cos x}{x}\, dxdy =$

(A) $\cos 1$.　　(B) $\sin 1$.

(C) $1-\cos 1$.　　(D) $1-\sin 1$.

2. (15 pts) Please fill in the blank for the questions below.

(1) If the function $z = z(x, y)$ is determined by $x^2 - 2y^2 + z^2 - 4x + 2z - 5 = 0$, then $\left.\frac{\partial z}{\partial y}\right|_{(5,2,2)} =$ ________________.

(2) $\lim\limits_{(x,y)\to(0,0)} \frac{\sin(xy^2)}{x^2+y^2} =$ ________________.

(3) If the region $D = \{(x, y)|\, x^2 + y^2 \le 1\}$, then $\iint_D e^{-x^2-y^2}\, dxdy =$ ________________.

(4) Let $\mathbf{F} = (z + e^{\sin y})\,\mathbf{i} + (\cos z - y)\,\mathbf{j} + (2z + \ln(1 + y^2))\,\mathbf{k}$. D is the upper semi-sphere $0 \le z \le \sqrt{a^2 - x^2 - y^2}\,(a \ge 0)$, and S is the boundary of the region D. Then the outward flux of $\mathbf{F}$ across S; i.e., $\iint_S \mathbf{F}\cdot\mathbf{n}\, d\sigma =$ ________________.

(5) $\int_C yze^{xz}dx + e^{xz}dy + xye^{xz}dz =$ ________________, where C is a path from $(2, 1, 0)$ to $(0, 4, 5)$.

3. (10 pts) Find the equation for the plane through the origin parallel to the following lines:

$$l_1 = \begin{cases} x = 1 \\ y = -1 + t \\ z = 2 + t \end{cases}, \quad l_2 = \begin{cases} x = -1 + t \\ y = -2 + 2t \\ z = 1 + t \end{cases}.$$

4. (10 pts) Use Taylor series to evaluate $\lim\limits_{n\to\infty} \left(n^3 \sin\frac{2}{n} - 2n^2\right)$.

5. (10 pts) In what directions is the directional derivative of $f(x, y) = xy + y^2$ at $P(3, 2)$ equal to zero ?

6. (10 pts) Compute $\iint_D xy\, dxdy$, here D is the disk enclosed by the curve $x^2 + y^2 = 2x + 2y$. (Hint: use substitution.)

7. (10 pts) Find the centroid of the region $D = \left\{(x, y, z)|\sqrt{x^2 + y^2} \le z \le \sqrt{1 - x^2 - y^2}\right\}$.

8. (10 pts) Calculate the line integral $\oint_C \mathbf{F}\cdot d\mathbf{r}$, where $\mathbf{F} = (y^2 + e^{e^x})\,\mathbf{i} + (xy + \cos y)\,\mathbf{j} + xz\,\mathbf{k}$, and C is the curve of intersection of the cylinder $x^2 + y^2 = 4y$ and the plane $y = z$, counterclockwise when viewed from above.

9. (10 pts) Find the absolute maximum and minimum values of the function $f(x, y) = 3x^2 + 4xy$ on the region R: $x^2 + y^2 \le 1$.

一、（15分）单项选择题:

(1) 幂级数 $\sum_{n=1}^{\infty}\frac{x^n}{n3^n}$ 的收敛域是

(A) $[-1/3,1/3]$.　　(B) $[-1/3,1/3)$.

(C) $[-3,3]$.　　(D) $[-3,3)$.

(2) 设 $f(x,y)=\begin{cases} y^2\sin\frac{1}{x^2+y^2}, & (x,y)\neq(0,0) \\ 0, & (x,y)=(0,0)\end{cases}$. 下列叙述中**错误**的是?

(A) $f(x,y)$ 在点 $(0,0)$ 处连续.

(B) $f_x(0,0)$ 存在.

(C) $f_y(0,0)$ 存在.

(D) $f_x(x,y)$ 在点 $(0,0)$ 处连续.

(3) 若函数 $f(x,y)$ 在点(x_0,y_0) 处的偏导数都存在. 则

(A) $f(x,y)$ 在 (x_0,y_0) 附近有界.

(B) $f(x,y)$ 在 (x_0,y_0) 附近连续.

(C) $f(x,y_0)$ 在 x_0 处连续, $f(x_0,y)$ 在y_0 处连续.

(D) $f(x,y)$ 在 (x_0,y_0) 处连续.

(4) 设 a 是一个常数，则级数 $\sum_{n=1}^{\infty}\left(\frac{\sin(an)}{n^2}+\frac{(-1)^n}{n+1}\right)$

(A) 绝对收敛.

(B) 条件收敛.

(C) 发散.

(D) 收敛性依赖于 a 的值.

(5) $\int_0^1\int_y^1\frac{\cos x}{x}\,dxdy=$

(A) $\cos 1$.　　(B) $\sin 1$.

(C) $1-\cos 1$.　　(D) $1-\sin 1$.

二、（15分）填空题:

(1) 设函数 $z=z(x,y)$ 由方程 $x^2-2y^2+z^2-4x+2z-5=0$ 所确定，则 $\left.\frac{\partial z}{\partial y}\right|_{(5,2,2)}=$ ________________.

(2) $\lim\limits_{(x,y)\to(0,0)}\frac{\sin(xy^2)}{x^2+y^2}=$ ________________.

(3) 设区域 $D=\{(x,y)|\,x^2+y^2\leq 1\}$，则 $\iint_D e^{-x^2-y^2}\,dxdy=$ ________________.

(4) 设向量场 $\mathbf{F}=(z+e^{\sin y})\,\mathbf{i}+(\cos z-y)\,\mathbf{j}+(2z+\ln(1+y^2))\,\mathbf{k}$. 区域 D 为上半球 $0\leq z\leq\sqrt{a^2-x^2-y^2}\,(a\geq 0)$，闭合曲面 S 是 D 的边界. 则向量场 $\mathbf{F}$ 通过曲面 S 从内向外的通量 $\iint_S\mathbf{F}\cdot\mathbf{n}\,d\sigma=$ ________________.

(5) $\int_C yze^{xz}dx+e^{xz}dy+xye^{xz}dz=$ ________________, 其中C为从$(2,1,0)$到$(0,4,5)$的一条路径.

三、（10分）求经过原点且平行于下面两条直线的平面方程

$$l_1=\begin{cases}x=1\\ y=-1+t\\ z=2+t\end{cases},\quad l_2=\begin{cases}x=-1+t\\ y=-2+2t\\ z=1+t\end{cases}.$$

四、（10分）使用泰勒级数来计算 $\lim\limits_{n\to\infty}\left(n^3\sin\dfrac{2}{n}-2n^2\right)$.

五、（10分）函数 $f(x,y)=xy+y^2$ 在 $P(3,2)$ 沿着哪些方向的方向导数为 0？

六、（10分）计算 $\iint_D xy\,dxdy$, 这里 D 是由闭合曲线 $x^2+y^2=2x+2y$ 围成的区域.（提示：用换元法）

七、（10分）求图形 D 的形心, 这里 D 是闭区域 $\left\{(x,y,z)|\sqrt{x^2+y^2}\le z\le\sqrt{1-x^2-y^2}\right\}$.

八、（10分）计算曲线积分 $\oint_C \mathbf{F}\cdot d\mathbf{r}$, 这里$\mathbf{F}=(y^2+e^{e^x})\,\mathbf{i}+(xy+\cos y)\,\mathbf{j}+xz\,\mathbf{k}$, 曲线$C$ 为圆柱面 $x^2+y^2=4y$ 与平面 $y=z$ 的交线，从上往下看， C 是逆时针方向.

九、（10分）求函数 $f(x,y)=3x^2+4xy$ 在闭区域 R: $x^2+y^2\le 1$ 的最大值和最小值（即全局极大和全局极小值）.

D 1. (15 pts) **Multiple Choice Questions:** (only one correct answer for each of the following questions.)

(1) The interval of convergence for the power series $\sum_{n=1}^{\infty} \frac{x^n}{n3^n}$ is

(A) $[-1/3, 1/3]$. (B) $[-1/3, 1/3)$.

(C) $[-3, 3]$. (D) $[-3, 3)$.

$$\lim_{n\to\infty} \sqrt[n]{\left|\frac{x^n}{n3^n}\right|} = \lim_{n\to\infty} \frac{1}{\sqrt[n]{n}} \frac{|x|}{3} = \frac{|x|}{3} < 1$$

$x \in (-3, 3)$ $\sum_{n=1}^{\infty} \frac{x^n}{n3^n}$ converges

$x = 3$ $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges

$x = -3$ $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ converges

D (2) Let $f(x, y) = \begin{cases} y^2 \sin \frac{1}{x^2+y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$. Which of the following statements is **wrong** ?

(A) $f(x, y)$ is continuous at $(0, 0)$.

(B) $f_x(0, 0)$ exists.

(C) $f_y(0, 0)$ exists.

(D) $f_x(x, y)$ is continuous at $(0, 0)$.

(A) $\left|y^2 \sin \frac{1}{x^2+y^2}\right| \le y^2$

$(x,y) \to (0,0)$ $y^2 \to 0$ $\left|y^2 \sin \frac{1}{x^2+y^2}\right| \to 0 = f(0,0)$

(B) $f_x(0,0) = \lim_{x\to 0} \frac{f(x,0) - f(0,0)}{x} = \lim_{x \to 0} \frac{0-0}{x} = 0$

(C) $f_y(0,0) = \lim_{y\to 0} \frac{f(0,y) - f(0,0)}{y} = \lim_{y\to 0} \frac{y^2 \sin \frac{1}{y^2}}{y} = \lim_{y\to 0} y \sin \frac{1}{y^2} = 0$

(D) $f_x(x,y) = y^2 \left(\cos \frac{1}{x^2+y^2}\right)\left(-\frac{2x}{(x^2+y^2)^2}\right) = -\frac{2xy^2}{(x^2+y^2)^2} \cos \frac{1}{x^2+y^2}$

$y = kx$ $f(x, kx) = -\frac{2k^2x^3}{(1+k^2)^2 x^4} \cos \frac{1}{(1+k^2)x^2}$

$= -\frac{2k^2}{(1+k^2)^2} \frac{1}{x} \cos \frac{1}{(1+k^2)x^2}$

$x_n = \frac{1}{\sqrt{1+k^2}\sqrt{2n\pi}}$ $\lim_{n\to\infty} f(x_n, kx_n) = \infty$

$x_n' = \frac{1}{\sqrt{1+k^2}\sqrt{2n\pi + \frac{\pi}{2}}}$ $\lim_{n\to\infty} f(x_n', kx_n') = 0$

$f_x(x,y)$ is not continuous at $(0,0)$.

C (3) If $f(x, y)$ has partial derivatives at (x_0, y_0), then

(A) $f(x, y)$ is bounded around (x_0, y_0).

(B) $f(x, y)$ is continuous around (x_0, y_0).

(C) $f(x, y_0)$ is continuous at x_0, $f(x_0, y)$ is continuous at y_0.

(D) $f(x, y)$ is continuous at (x_0, y_0).

$$f_x(x_0,y_0) = \lim_{(x,y_0)\to(x_0,y_0)} \frac{f(x,y_0)-f(x_0,y_0)}{x-x_0} \quad \text{exists}$$

$$f_y(x_0,y_0) = \lim_{(x_0,y)\to(x_0,y_0)} \frac{f(x_0,y)-f(x_0,y_0)}{y-y_0} \quad \text{exists.}$$

B (4) Let a be a constant. Then the series $\sum_{n=1}^{\infty} \left(\frac{\sin(an)}{n^2} + \frac{(-1)^n}{n+1} \right)$

(A) converges absolutely.

(B) converges conditionally.

(C) diverges.

(D) the convergence depends on the value of a.

$$\left|\frac{\sin an}{n^2}\right| \le \frac{1}{n^2} \qquad \sum_{n=1}^{\infty} \frac{1}{n^2} \text{ converges}$$

Comparision Test, $\sum_{n=1}^{\infty} \frac{\sin(an)}{n^2}$ converges absolutely

$\sum_{n=1}^{\infty} \frac{(-1)^n}{n+1}$ converges $\qquad \sum_{n=1}^{\infty} \frac{1}{n+1}$ diverges

$\sum_{n=1}^{\infty} \frac{\sin an}{n^2} + \frac{(-1)^n}{n+1}$ converges conditionally.

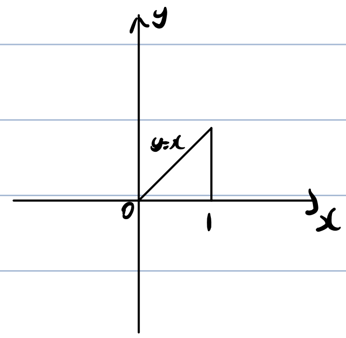
B (5) $\int_0^1 \int_y^1 \frac{\cos x}{x}\, dx dy =$

(A) $\cos 1$.

(B) $\sin 1$.

(C) $1 - \cos 1$.

(D) $1 - \sin 1$.



$$\int_0^1 \int_y^1 \frac{\cos x}{x}\, dx dy = \int_0^1 \int_0^x \frac{\cos x}{x}\, dy dx$$

$$= \int_0^1 x \cdot \frac{\cos x}{x}\, dx = \int_0^1 \cos x\, dx = \sin x \Big|_0^1 = \sin 1$$

2. (15 pts) Please fill in the blank for the questions below.

(1) If the function $z = z(x, y)$ is determined by $x^2 - 2y^2 + z^2 - 4x + 2z - 5 = 0$, then $\frac{\partial z}{\partial y}\Big|_{(5,2,2)} =$ $\frac{4}{3}$.

$$-4y + 2z\frac{\partial z}{\partial y} + 2\frac{\partial z}{\partial y} = 0 \qquad \frac{\partial z}{\partial y} = \frac{2y}{z+1} \qquad \frac{\partial z}{\partial y}\Big|_{(5,2,2)} = \frac{2\cdot 2}{2+1} = \frac{4}{3}$$

(2) $\lim_{(x,y)\to(0,0)} \dfrac{\sin(xy^2)}{x^2+y^2} =$ 0.

$$\left|\frac{\sin xy^2}{x^2+y^2}\right| \le \left|\frac{xy^2}{x^2+y^2}\right| \le \left|\frac{\frac{1}{2}(x^2+y^2)y}{x^2+y^2}\right| = \left|\frac{y}{2}\right|$$

$(x,y)\to(0,0) \quad \left|\frac{y}{2}\right|\to 0$

$$\left|\frac{\sin xy^2}{x^2+y^2}\right| \to 0$$

(3) If the region $D = \{(x,y) | x^2+y^2 \le 1\}$, then $\iint_D e^{-x^2-y^2}\,dxdy =$ $\pi(1-\frac{1}{e})$.

$$\iint_D e^{-(x^2+y^2)}\,dxdy \overset{\substack{x=r\cos\theta \\ y=r\sin\theta}}{=} \int_0^{2\pi}\int_0^1 e^{-r^2} r\,dr\,d\theta = \int_0^{2\pi} -\frac{1}{2}e^{-r^2}\Big|_0^1\,d\theta = 2\pi\cdot\left(-\frac{1}{2e}+\frac{1}{2}\right)$$

$$= \pi\left(1-\frac{1}{e}\right)$$

(4) Let $\mathbf{F} = (z+e^{\sin y})\,\mathbf{i} + (\cos z - y)\,\mathbf{j} + (2z + \ln(1+y^2))\,\mathbf{k}$. D is the upper semi-sphere $0 \le z \le \sqrt{a^2-x^2-y^2}\,(a \ge 0)$, and S is the boundary of the region D. Then the outward flux of $\mathbf{F}$ across S; i.e., $\iint_S \mathbf{F}\cdot\mathbf{n}\,d\sigma =$ $\frac{2\pi a^3}{3}$.

$$\iint_S F\cdot n\,d\sigma = \iiint_D \nabla\cdot F\,dV = \iiint_D 1\,dV = \frac{1}{2}\cdot\frac{4}{3}\pi a^3 = \frac{2\pi a^3}{3}$$

$$\nabla\cdot F = \frac{\partial(z+e^{\sin y})}{\partial x} + \frac{\partial(\cos z - y)}{\partial y} + \frac{\partial(2z+\ln(1+y^2))}{\partial z} = 1$$

(5) $\int_C yze^{xz}dx + e^{xz}dy + xye^{xz}dz =$ 3, where C is a path from $(2,1,0)$ to $(0,4,5)$.

$M = yze^{xz} \quad N = e^{xz} \quad P = xye^{xz}$

$$\frac{\partial M}{\partial y} = ze^{xz} = \frac{\partial N}{\partial x} \qquad \frac{\partial N}{\partial z} = xe^{xz} = \frac{\partial P}{\partial y} \qquad \frac{\partial M}{\partial z} = ye^{xz} + xyze^{xz} = \frac{\partial P}{\partial x}$$

$\exists f \qquad df = Mi + Nj + Pk$

$f = ye^{xz} + C$

$$\int_C df = f(0,4,5) - f(2,1,0) = 4e^0 + C - e^0 - C = 3$$

3. (10 pts) Find the equation for the plane through the origin parallel to the following lines:

$$l_1 = \begin{cases} x = 1 \\ y = -1+t \\ z = 2+t \end{cases}, \quad l_2 = \begin{cases} x = -1+t \\ y = -2+2t \\ z = 1+t \end{cases}.$$

The direction of l_1 : $V_1 = (0, 1, 1)$

The direction of l_2 : $V_2 = (1, 2, 1)$

$$\vec{n} = V_1 \times V_2 = \begin{vmatrix} i & j & k \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{vmatrix} = -i + j - k$$

The equation of the plane :

$$-(x-0) + (y-0) - (z-0) = 0$$

$$x - y + z = 0.$$

4. (10 pts) Use Taylor series to evaluate $\lim_{n\to\infty}\left(n^3 \sin\frac{2}{n} - 2n^2\right)$.

$$\lim_{n\to\infty} \left(n^3 \sin\frac{2}{n} - 2n^2\right)$$

$$= \lim_{n\to\infty} \left(n^3\left(\frac{2}{n} - \frac{2^3}{3!\,n^3} + o\left(\left(\frac{2}{n}\right)^3\right)\right) - 2n^2\right)$$

$$= \lim_{n\to\infty} \left(2n^2 - \frac{4}{3} + o(1) - 2n^2\right)$$

$$= \lim_{n\to\infty} \left(-\frac{4}{3} + o(1)\right) = -\frac{4}{3}.$$

5. (10 pts) In what directions is the directional derivative of $f(x,y) = xy + y^2$ at $P(3,2)$ equal to zero ?

$$\nabla f = \frac{\partial f}{\partial x} i + \frac{\partial f}{\partial y} j$$

$$= y\, i + (x+2y) j$$

$$(\nabla f)_p = 2i + 7j \qquad u = \cos\theta\, i + \sin\theta\, j$$

$$\left(\frac{df}{ds}\right)_{u,p} = (2i+7j)(\cos\theta\, i + \sin\theta\, j) = 2\cos\theta + 7\sin\theta = 0$$

$$\tan\theta = -\frac{2}{7} \Rightarrow ① \ \sin\theta = \frac{2\sqrt{53}}{53} \quad \cos\theta = -\frac{7\sqrt{53}}{53} \Rightarrow u_1 = \frac{2\sqrt{53}}{53} i - \frac{7\sqrt{53}}{53} j$$

$$② \ \sin\theta = -\frac{2\sqrt{53}}{53} \quad \cos\theta = \frac{7\sqrt{53}}{53} \qquad u_2 = -\frac{2\sqrt{53}}{53} i + \frac{7\sqrt{53}}{53} j$$

6. (10 pts) Compute $\iint_D xy\,dxdy$, here D is the disk enclosed by the curve $x^2+y^2=2x+2y$.
(Hint: use substitution.)

$$x^2+y^2=2x+2y \Rightarrow (x-1)^2+(y-1)^2=2 \Rightarrow D=\{(x,y)\mid (x-1)^2+(y-1)^2\le 2\}$$

$$\text{Let } \begin{cases} x = r\cos\theta+1 \\ y = r\sin\theta+1 \end{cases} \qquad \frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r$$

$$\iint_D xy\,dxdy = \int_0^{2\pi}\int_0^{\sqrt{2}} (r\cos\theta+1)(r\sin\theta+1)\,r\,dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^{\sqrt{2}} \left(r^3\sin\theta\cos\theta + r^2\cos\theta + r^2\sin\theta + r\right) dr\,d\theta$$

$$= \int_0^{2\pi}\sin\theta\cos\theta\,d\theta\int_0^{\sqrt{2}} r^3dr + \int_0^{2\pi}\cos\theta\,d\theta\int_0^{\sqrt{2}} r^2dr + \int_0^{2\pi}\sin\theta\,d\theta\int_0^{\sqrt{2}} r^2dr$$

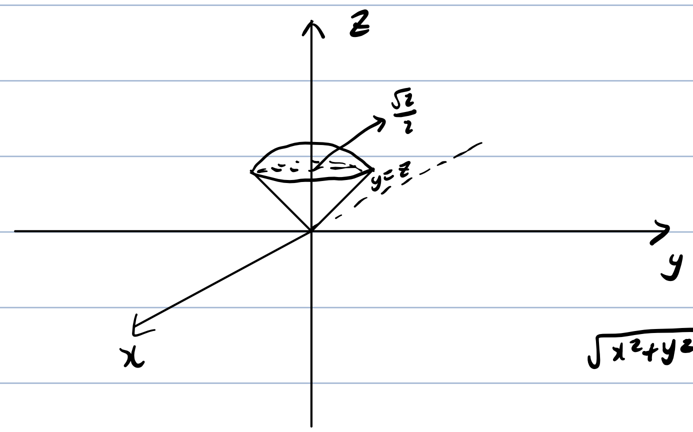
$$+ \int_0^{2\pi}d\theta\int_0^{\sqrt{2}} r\,dr$$

$$= \int_0^{2\pi}\frac{1}{2}\sin2\theta\,d\theta \;\frac{1}{4}r^4\Big|_0^{\sqrt{2}} + \sin\theta\Big|_0^{2\pi}\frac{1}{3}r^3dr - \cos\theta\Big|_0^{2\pi}\frac{1}{3}r^3\Big|_0^{\sqrt{2}} + 2\pi\,\frac{1}{2}r^2\Big|_0^{\sqrt{2}}$$

$$= -\frac{1}{4}\cos2\theta\Big|_0^{2\pi}\frac{1}{4}\cdot 4 + 0 - 0 + 2\pi$$

$$= 0 + 2\pi = 2\pi$$

7. (10 pts) Find the centroid of the region $D=\left\{(x,y,z)\middle|\sqrt{x^2+y^2}\le z\le\sqrt{1-x^2-y^2}\right\}$.



$$\text{Let } \begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ z = z \end{cases}$$

$$\sqrt{x^2+y^2} = \sqrt{1-x^2+y^2} \Rightarrow x^2+y^2=\frac{1}{2}$$

$$M = \iiint_D dV = \int_0^{2\pi}\int_0^{\frac{\sqrt{2}}{2}}\int_r^{\sqrt{1-r^2}} r\,dz\,dr\,d\theta = \int_0^{2\pi}\int_0^{\frac{\sqrt{2}}{2}}\left(r\sqrt{1-r^2}-r^2\right)dr\,d\theta$$

$$= \int_0^{2\pi} \left(-\frac{1}{3}(1-r^2)^{\frac{3}{2}} - \frac{1}{3}r^3\right)\Big|_0^{\frac{\sqrt{2}}{2}} d\theta$$

$$= 2\pi\left(-\frac{1}{3}\cdot\frac{\sqrt{2}}{4} - \frac{1}{3}\cdot\frac{\sqrt{2}}{4} + \frac{1}{3}\right)$$

$$= \frac{(2-\sqrt{2})\pi}{3}$$

$$M_{yz} = \iiint_D x\,dV = \int_0^{2\pi}\int_0^{\frac{\sqrt{2}}{2}}\int_r^{\sqrt{1-r^2}} r^2\cos\theta\, dz\,dr\,d\theta$$

$$= \int_0^{2\pi}\cos\theta\,d\theta \int_0^{\frac{\sqrt{2}}{2}}\int_r^{\sqrt{1-r^2}} r^2\,dz\,dr = 0$$

$$M_{zx} = \iiint_D y\,dV = \int_0^{2\pi}\int_0^{\frac{\sqrt{2}}{2}}\int_r^{\sqrt{1-r^2}} r^2\sin\theta\,d\theta\,dr\,d\theta = \int_0^{2\pi}\sin\theta\,d\theta\int_0^{\frac{\sqrt{2}}{2}}\int_r^{\sqrt{1-r^2}} r^2\,dz\,dr = 0$$

$$M_{xy} = \iiint_D z\,dV = \int_0^{2\pi}\int_0^{\frac{\sqrt{2}}{2}}\int_r^{\sqrt{1-r^2}} rz\,dz\,dr\,d\theta = \int_0^{2\pi}\int_0^{\frac{\sqrt{2}}{2}} \frac{1}{2}z^2 r\Big|_r^{\sqrt{1-r^2}} dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^{\frac{\sqrt{2}}{2}} \frac{1}{2} r\sqrt{1-2r^2}\,dr\,d\theta$$

$$= \int_0^{2\pi} -\frac{1}{12}(1-2r^2)^{\frac{3}{2}}\Big|_0^{\frac{\sqrt{2}}{2}} d\theta$$

$$= 2\pi\cdot\left(\frac{1}{12}\right) = \frac{\pi}{6}$$

$$\bar{x} = \frac{M_{yz}}{M} = 0 \qquad \bar{y} = \frac{M_{zx}}{M} = 0 \qquad \bar{z} = \frac{M_{xy}}{M} = \frac{\frac{\pi}{6}}{\frac{(2-\sqrt{2})\pi}{3}} = \frac{2+\sqrt{2}}{4}$$

Centroid $\left(0, 0, \frac{2+\sqrt{2}}{4}\right)$

8. (10 pts) Calculate the line integral $\oint_C \mathbf{F}\cdot d\mathbf{r}$, where $\mathbf{F} = (y^2 + e^{e^x})\,\mathbf{i} + (xy + \cos y)\,\mathbf{j} + xz\,\mathbf{k}$, and C is the curve of intersection of the cylinder $x^2 + y^2 = 4y$ and the plane $y = z$, counterclockwise when viewed from above.

$$x^2+y^2=4y \Rightarrow x^2+(y-2)^2=4$$

Let $\begin{cases} x = 2\cos\theta \\ y = 2+2\sin\theta \\ z = z \end{cases}$

S is the region enclosed by C.

$$S = \left\{ (x,y,z) \,\middle|\, \begin{matrix} x^2+(y-2)^2 \le 4 \\ y = z \end{matrix} \right\} = \left\{ (r,\theta) \,\middle|\, \begin{matrix} x = r\cos\theta & y = 2+r\sin\theta & z = 2+r\sin\theta \\ 0 \le r \le 2 & & 0 \le \theta \le 2\pi \end{matrix} \right\}.$$

$$\nabla\times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2+e^{e^x} & xy+\cos y & xz \end{vmatrix} = -zj + 3yk$$

S: $h(r,\theta) = r\cos\theta\, i + (2+r\sin\theta) j + (2+r\sin\theta)k$

$h_r = \cos\theta\, i + \sin\theta\, j + \sin\theta\, k$

$h_\theta = -r\sin\theta\, i + r\cos\theta\, j + r\cos\theta\, k$

$$n = \frac{h_r\times h_\theta}{|h_r\times r_\theta|} = \frac{1}{|h_r\times h_\theta|}\begin{vmatrix} i & j & k \\ \cos\theta & \sin\theta & \sin\theta \\ -r\sin\theta & r\cos\theta & r\cos\theta \end{vmatrix} = \frac{1}{\sqrt{(-r)^2+r^2}}(-rj+rk) = -\frac{\sqrt{2}}{2}j + \frac{\sqrt{2}}{2}k$$

$$\nabla\times F\cdot n|_S = \left(\frac{\sqrt{2}z}{2} + \frac{3\sqrt{2}y}{2}\right)\Big|_S = 2\sqrt{2}(2+r\sin\theta)$$

$$d\sigma = |h_r\times h_\theta|\,dr\,d\theta = \sqrt{2}\,r\,dr\,d\theta$$

By Stoke's Theorem,

$$\oint_C F\cdot dr = \iint_S \nabla\times F\cdot n\, d\sigma = \int_0^{2\pi}\int_0^2 2\sqrt{2}(2+r\sin\theta)\sqrt{2}\,r\,dr\,d\theta$$

$$= 4\int_0^{2\pi}\int_0^2 (2r + r^2\sin\theta)\,dr\,d\theta$$

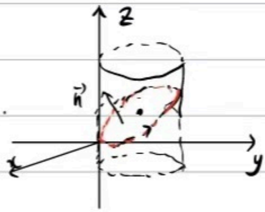
$$= 4\int_0^{2\pi} d\theta \int_0^2 2r\,dr + 4\int_0^{2\pi}\sin\theta\, d\theta \int_0^2 r^2\,dr$$

$$= 4\cdot 2\pi\cdot r^2\Big|_0^2 + 4(-\cos\theta)\Big|_0^{2\pi}\, \frac{1}{3}r^3\Big|_0^2$$

$$= 32\pi$$

$x^2+y^2=4y \Rightarrow x^2+(y-2)^2=4$

S is the region enclosed by C.



$$\nabla\times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2+e^{e^x} & xy+\cos y & xz \end{vmatrix} = -zj - yk \qquad \vec{n} = -\frac{\sqrt{2}}{2}(j-k) = -\frac{\sqrt{2}}{2}j+\frac{\sqrt{2}}{2}k$$

$$\nabla\times F\cdot n\Big|_S = \left(\frac{\sqrt{2}z}{2}-\frac{\sqrt{2}y}{2}\right)\Big|_S = 0$$

By Stoke's Theorem,

$$\oint_C F\cdot dr = \iint_S \nabla\times F\cdot n\, d\sigma = \iint_S 0\, d\sigma = 0$$

9. (10 pts) Find the absolute maximum and minimum values of the function $f(x,y) = 3x^2 + 4xy$ on the region R: $x^2 + y^2 \le 1$.

Consider $x^2+y^2<1$

$f_x = 6x+4y$ $\quad \begin{cases} f_x = 0 \\ f_y = 0 \end{cases} \Rightarrow \begin{cases} x=0 \\ y=0 \end{cases} \Rightarrow f(0,0)=0$

$f_y = 4x$

Consider $x^2+y^2=1$

Let $g(x,y) = x^2+y^2-1.$

$\nabla f = (6x+4y)\vec{i} + 4x\vec{j}$

$\nabla g = 2x\vec{i} + 2y\vec{j}$

$\nabla f = \lambda \nabla g \Rightarrow \begin{cases} 6x+4y = 2\lambda x \\ 4x = 2\lambda y \end{cases}$

$\Rightarrow (3-\lambda)x + 2y = 0$

$(3-\lambda)\frac{\lambda y}{2} + 2y = 0$

$(3\lambda - \lambda^2 + 4)y = 0$

$\Rightarrow \lambda_1 = 4 \quad \lambda_2 = -1 \quad y=0$

① $\lambda_1 = 4$ $\quad x^2+y^2=1 \Rightarrow \begin{cases} x = \frac{2\sqrt5}{5} \\ y = \frac{\sqrt5}{5} \end{cases} \begin{cases} x = -\frac{2\sqrt5}{5} \\ y = \frac{\sqrt5}{5} \end{cases}$ $\quad f(\frac{2\sqrt5}{5}, \frac{\sqrt5}{5}) = f(-\frac{2\sqrt5}{5}, \frac{\sqrt5}{5})$

$x = 2y$

$= 3\cdot\frac{4}{5} + 4\cdot\frac{2}{5} = 4$

② $\lambda_2 = 1$ $\quad x^2+y^2=1 \Rightarrow \begin{cases} x = \frac{\sqrt5}{5} \\ y = -\frac{2\sqrt5}{5} \end{cases} \begin{cases} x = -\frac{\sqrt5}{5} \\ y = \frac{2\sqrt5}{5} \end{cases}$ $\quad f(\frac{\sqrt5}{5}, -\frac{2\sqrt5}{5}) = f(-\frac{\sqrt5}{5}, \frac{2\sqrt5}{5})$

$-2x = y$

$= 3\cdot\frac{1}{5} + 4(-\frac{2}{5}) = -1$

③ $y=0$ $\quad x=0$ $\quad (x,y) \in \{(x,y) \mid x^2+y^2=1\} \Rightarrow$ Contradiction.

$f_{absmax} = f(\frac{2\sqrt5}{5}, \frac{\sqrt5}{5}) = f(-\frac{2\sqrt5}{5}, \frac{\sqrt5}{5}) = 4$

$f_{absmin} = f(\frac{\sqrt5}{5}, -\frac{2\sqrt5}{5}) = f(-\frac{\sqrt5}{5}, \frac{2\sqrt5}{5}) = -1$